

Rohit Anil Shinde

Machine Learning Engineer | AI/ML Systems Developer

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Immediate Availability

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Profile | Kurzprofil

Machine Learning Engineer with 4+ years of hands-on experience designing, training, and deploying production AI systems across the complete software lifecycle. Strong foundation in statistical modeling, NLP, computer vision, and deep learning. Specialized in fine-tuning open-weights LLMs (QLoRA, PEFT), curriculum-grounded RAG pipelines, and containerized low-latency microservices with Python, FastAPI, Docker, and GCP.

Beruflicher Werdegang | Professional Experience

12/2025 – 06/2026	AI / GenAI Engineer — Machine Learning & AI Systems EduAI Hub <i>Pune, India</i>
	■ Led development of a domain-specific ML and language-model system; fine-tuned Gemma 3B using LoRA, QLoRA, and progressive layer unfreezing, improving domain-specific Q&A accuracy by approximately 38%. ■ Designed production ML pipelines covering data preparation, model training, evaluation, experiment tracking, and deployment. ■ Architected retrieval and data-processing pipelines using LlamaIndex and ChromaDB, grounding outputs in curriculum documents and reducing hallucinations by approximately 42%. ■ Optimized model inference and serving infrastructure, reducing latency by approximately 40% through parameter-efficient techniques and vLLM-based serving. ■ Established MLOps workflows using MLflow and Weights & Biases for experiment tracking, model evaluation, versioning, and reproducibility. ■ Developed and deployed production AI/ML microservices using Python, FastAPI, Docker, and GCP Cloud Run, supported by CI/CD pipelines. ■ Designed multimodal ML workflows using Stable Diffusion, SDXL, and ControlNet for automated educational content generation. ■ Mentored junior engineers on ML workflows, model development, RAG systems, and MLOps practices.
01/2024 – Present	Freelance ML / AI Engineer Independent Projects <i>Remote</i>
	■ Delivered end-to-end ML, NLP and intelligent automation systems for international clients from discovery through production deployment. ■ Engineered Python data-processing pipelines, reducing operational data-handling overhead by 70%. ■ Built production REST APIs and asynchronous inference endpoints using FastAPI and Celery.
06/2021 – Present	Founder & Lead Engineer Stones Web Services <i>Pune, India</i>
	■ Founded a digital solutions agency delivering cloud architectures and ML integrations for 15+ startups and regional businesses. ■ Built production web apps and APIs with Python (FastAPI, Django) and React, integrating NLP and computer-vision modules. ■ Managed containerized GCP infrastructure, deployment workflows and automated CI/CD.
11/2024 – 04/2025	Machine Learning Researcher / Developer Dr. D. Y. Patil SOST <i>Pune, India</i>
	■ Researched and implemented Machine Learning, Computer Vision, and NLP techniques for an AI-powered electronic voting and identity-verification system. ■ Developed ML-based fraud and anomaly-detection capabilities to identify potentially suspicious voting activity. ■ Researched computer-vision and facial-recognition approaches for identity verification and system security. ■ Investigated diffusion-based synthetic data generation for security-model training. ■ Contributed to system architecture spanning machine learning, analytics, identity verification, and security components.
08/2024 – 11/2024	Technical Trainer (Python & Web Frameworks) Teknowell EduTech <i>Pune, India</i>
	■ Trained 50+ students in Python, Django, Flask, REST APIs, React, Git, deployment, and software development practices. ■ Mentored students on project development, debugging, deployment, and technical problem solving. ■ Explained complex technical concepts to learners with different levels of programming experience.

Ausgewählte Projekte | Selected Projects

2025 – 2026	EduAI SLM — Machine Learning Model Fine-Tuning & RAG EduAI Hub <i>Production</i>
Stack: PyTorch • Hugging Face • Scikit-learn • LlamaIndex • ChromaDB • FastAPI • GCP • MLflow	
■ Fine-tuned Gemma 3B using LoRA, QLoRA, and progressive layer unfreezing for domain-specific machine learning applications. ■ Improved domain-specific Q&A accuracy by approximately 38% through model adaptation and evaluation. ■ Built data ingestion and retrieval pipelines using LlamaIndex and ChromaDB. ■ Reduced hallucinations by approximately 42% through retrieval-grounded model responses. ■ Implemented production inference and serving on GCP Cloud Run and established ML experiment tracking using MLflow.	
2024 – 2025	AI-Powered Electronic Voting System Research Initiative <i>Prototype</i>
Stack: Python • Django • FastAPI • Computer Vision • Machine Learning	
■ Developed an electronic voting platform incorporating facial recognition, two-factor authentication, real-time analytics, and ML-based fraud detection. ■ Applied computer vision and machine learning techniques for identity verification and anomaly detection. ■ Designed Django and FastAPI backend services to integrate ML capabilities with production application workflows.	
2025 – 2026	Multi-Modal ML Content Pipeline Stones Web Services <i>Production</i>
Stack: PyTorch • Stable Diffusion • SDXL • ControlNet • LoRA • FastAPI	
■ Developed a machine learning pipeline for automated educational diagram and visual-content generation. ■ Integrated SDXL, ControlNet, and LoRA for controlled image-generation workflows. ■ Exposed ML workflows through FastAPI microservices and designed the pipeline for Learning Management System integration.	

Ausbildung | Education

08/2021 – 06/2025	Bachelor of Technology (B.Tech) in AI & Data Science Dr. D. Y. Patil Vidyapeeth <i>Pune, India</i>
CGPA: 9.45 / 10.0 German Equivalent: $\approx$ 1,1 (Modified Bavarian Formula)	
■ Key Modules: Machine Learning, Deep Neural Networks, Probability & Applied Statistics, Computer Vision, Natural Language Processing, Data Structures & Algorithms.	

Fachliche Kompetenzen | Technical Competencies

Programming	Python (Core), SQL, TypeScript, JavaScript, C++, Bash
ML & Math	PyTorch, TensorFlow, Scikit-learn, Hugging Face, SciPy, NumPy, Pandas, Statistical Modeling
GenAI & LLMs	RAG, Fine-Tuning (LoRA, QLoRA, PEFT), vLLM, Prompt Engineering, LlamaIndex, LangChain
Vector Search	ChromaDB, FAISS, Milvus, Qdrant, Semantic Embeddings, Hierarchical Chunking
MLOps & Cloud	Docker, GCP (Vertex AI, Cloud Run, GCS), MLflow, W&B, Git, GitHub Actions (CI/CD)
Backend	FastAPI, Django, Flask, asyncio, REST APIs, Celery, Redis, PostgreSQL

Zertifikate | Certifications

Google	Google Advanced Data Analytics Professional Certificate (Statistical Modeling & Predictive ML)
Coursera	The Power of Statistics: Inference, Hypothesis Testing & Mathematical Modeling

Sprachkenntnisse & Status | Languages & Mobility

Languages	English (Full Professional Proficiency, C1), German (Elementary, A1 in progress), Hindi (Bilingual), Marathi (Native)
Work Authorization	Indian Citizen Immediate Availability Eligible for German EU Blue Card / Fast-Track Skilled Worker Visa